Pupil Name: _____ **Class:** _____

Medicine in Britain, c1250–present with Western Front

PEARSON EDEXCEL GCSE (9–1) HISTORY • PAPER 1: THEMATIC STUDY & HISTORIC ENVIRONMENT (1HI0/11) • MASTER KNOWLEDGE RETRIEVAL & HOMEWORK COMPANION



COGNITIVE SCIENCE **WHY RETRIEVAL PRACTICE WORKS • OVERCOMING THE FORGETTING CURVE**

Human memory decays exponentially after learning—a reality proven by Hermann Ebbinghaus's *Forgetting Curve*. While passive re-reading creates a deceptive illusion of mastery, **deliberate retrieval practice** actively reconstructs neural pathways, arresting forgetfulness and cementing essential historical knowledge into long-term schema.

Archival Plate: *Source A: Royal Army Medical Corps stretcher-bearers carrying a wounded soldier through knee-deep mud near Boesinghe during the Third Battle of Ypres (Passchendaele, 1 August 1917). Imperial War Museum (IWM Q 5935).*

SPECIFICATION REFERENCE & HISTORICAL FOCUS	HW DUE DATE	1. SOLO (WRITTEN)	2. PARTNER (VERBAL)	PARENT / TEACHER SIGN
KT 1.1: Medieval Ideas on Disease: Religion, Humours & Miasma (c1250–c1500) God's punishment, astrology (1345 conjunction), Theory of Four Humours, opposites, Galen & miasma	_____	____ / 16	____ / 16	_____
KT 1.2: Medieval Care, Treatments & The 1348 Black Death Phlebotomy, purging, herbal remedies, wise women, monastic hospitals, and the Black Death (1348–1349)	_____	____ / 16	____ / 16	_____
KT 2.1: The Medical Renaissance: Vesalius, Sydenham & The Royal Society Humanism, movable type printing press (1440), Andreas Vesalius (1543), Thomas Sydenham (1676) & Royal Society (1660)	_____	____ / 16	____ / 16	_____
KT 2.2: William Harvey's Circulation & The 1665 Great Plague William Harvey (1628), mechanical pump heart, one-way valves, continuity in treatments & Great Plague of London	_____	____ / 16	____ / 16	_____
KT 3.1: Germ Theory & Microbe Hunters (c1861–c1883) Spontaneous generation, Louis Pasteur (1861 Germ Theory), swan-neck flasks, Robert Koch, agar jelly & specific microbes	_____	____ / 16	____ / 16	_____
KT 3.2: Prevention & Hospital Care: Smallpox Vaccine & Nightingale Inoculation, Edward Jenner (1796), cowpox, James Phipps, Florence Nightingale, pavilion hospitals & Mary Seacole	_____	____ / 16	____ / 16	_____
KT 3.3: The Surgical Revolution & Public Health (1847–1875) Humphry Davy, James Simpson (1847 Chloroform), Joseph Lister (1865 Carbolic Acid), John Snow & 1875 Public Health Act	_____	____ / 16	____ / 16	_____
KT 4.1: Modern Medicine: Magic Bullets, Penicillin & The NHS Paul Ehrlich (Salvarsan 606), Alexander Fleming (1928), Florey & Chain (1940), US mass production & NHS (1948)	_____	____ / 16	____ / 16	_____
KT 5.1: The Western Front: Trench Geography & Environmental Illness Ypres Salient, Hill 60, Somme, Arras underground tunnels, Cambrai, trench foot, trench fever & shell shock	_____	____ / 16	____ / 16	_____
KT 5.2: Battlefield Trauma & Weapons: Shrapnel, Gas & Gangrene Artillery shrapnel (58% wounds), high explosives, gas gangrene in fertilised soil, chlorine, phosgene & mustard gas	_____	____ / 16	____ / 16	_____
KT 5.3: The Chain of Evacuation: RAP to Base Hospitals Stretcher Bearers, RAP (Regimental Aid Post), ADS & MDS Dressing Stations, CCS (Casualty Clearing Station) & Base Hospital	_____	____ / 16	____ / 16	_____
KT 5.4: Surgical Breakthroughs: Thomas Splint, X-Rays & Blood Storage Wound debridement, Carrel-Dakin method, Thomas Splint, mobile X-ray vans, Richard Lewisohn & 1917 Cambrai blood bank	_____	____ / 16	____ / 16	_____

HOW TO COMPLETE YOUR RETRIEVAL CYCLES:

1. SOLO (WRITTEN): Complete each 8-question enquiry from memory in test conditions. Every question requires a precise **Core Fact** and **Historical Explanation**. Self-mark and correct in green pen using the Department Answer Bank (pp. 14–16).

2. PARTNER (VERBAL): Hand your booklet to a **parent, carer, family member, or study buddy** to test you out loud from the Answer Bank until you achieve 16/16 mastery!

KT 1.1: Medieval Ideas on Disease: Religion, Humours & Miasma (c1250–c1500)

Key Enquiry Question: "How did religious beliefs, the Four Humours, and miasma shape medieval ideas about disease (c1250–c1500)?"

CHRONOLOGICAL SPINE

- c350 BC — Hippocrates' Four Humours
- → c170 AD — Galen's Theory of Opposi...
- → c1250 — Church Controls Universit...
- → Mar 1345 — Conjunction of 3 Planets
- → c1400 — Urine Wheel Diagnosis St...

Q1 What did medieval people believe was the primary supernatural cause of disease? / 2m

CORE FACT:

EXPLANATION:

Q2 How did the medieval Catholic Church control medical ideas and university training? / 2m

CORE FACT:

EXPLANATION:

Q3 In March 1345, what unusual alignment of planets did medieval scholars blame for causing disease? / 2m

CORE FACT:

EXPLANATION:

Q4 What was the idea of "miasma" in medieval medical thinking? / 2m

CORE FACT:

EXPLANATION:

Q5 What religious penance was carried out by groups of flagellants during deadly plagues? / 2m

CORE FACT:

EXPLANATION:

Q6 Name the Four Humours identified by ancient Greek doctor Hippocrates. / 2m

CORE FACT:

EXPLANATION:

Q7 Which qualities and season were linked with the humour of Blood? / 2m

CORE FACT:

EXPLANATION:

Q8 In the 2nd century AD, what theory did Roman doctor Galen create to balance the humours? / 2m

CORE FACT:

EXPLANATION:

⚠ **EXAMINER WARNING & FATAL PITFALL**

Fatal Examiner Trap: Believing medieval people only blamed God for disease. While supernatural beliefs were dominant, rational explanations (Hippocrates' Four Humours, Galen's Theory of Opposites, and miasma) were taught in universities and used by physicians.

KT 1.2: Medieval Care, Treatments & The 1348 Black Death

Key Enquiry Question: "How effective were medieval medical treatments, care, and responses to the 1348 Black Death?"

CHRONOLOGICAL SPINE

c1200 Monastic Hospitals Care ... → 1348 Plague Enters Melcombe ... → 1348 Edward III Orders Street C... → 1349 50% of English Populatio... → 1350 Labour Shortage Transfor...

Q1 What was the most common humoural treatment used by medieval doctors to balance the body? / 2m

CORE FACT:

EXPLANATION:

Q2 Name the three main practical ways medieval healers took blood from patients. / 2m

CORE FACT:

EXPLANATION:

Q3 What treatments were given to clear bad humours out of the stomach and bowels? / 2m

CORE FACT:

EXPLANATION:

Q4 What ancient health guide gave personalised advice on diet, rest, exercise, and the Six Non-Naturals? / 2m

CORE FACT:

EXPLANATION:

Q5 What phrase describes the main goal of medieval English hospitals, which offered shelter rather than medicine? / 2m

CORE FACT:

EXPLANATION:

Q6 Who staffed and funded the majority of the 500+ hospitals in England by 1400? / 2m

CORE FACT:

EXPLANATION:

Q7 Which medical practitioner had a university degree, charged high fees, and diagnosed illnesses? / 2m

CORE FACT:

EXPLANATION:

Q8 In 1348, the Black Death arrived in England at Melcombe Regis. Roughly what fraction of the population died? / 2m

CORE FACT:

EXPLANATION:

⚠ **EXAMINER WARNING & FATAL PITFALL**

Fatal Examiner Trap: Believing medieval hospitals cured disease. Medieval hospitals provided "care, not cure" (food, shelter, warmth, and prayer run by monks and nuns); infectious sufferers and pregnant women were turned away.

KT 2.1: The Medical Renaissance: Vesalius, Sydenham & The Royal Society

Key Enquiry Question: "How did the anatomical work of Vesalius and Sydenham challenge traditional medical ideas in the Renaissance?"

CHRONOLOGICAL SPINE

1543 Vesalius Publishes Fabric → 1560 Paré's Soothing Wound B... → 1660 Royal Society Founded → 1665 Great Plague Hits London → 1676 Sydenham's Medical Obs...

Q1 What Renaissance movement encouraged scholars to question ancient ideas and observe nature directly? / 2m

CORE FACT:

EXPLANATION:

Q2 Around 1440, what invention allowed medical books and ideas to be shared quickly without copying errors? / 2m

CORE FACT:

EXPLANATION:

Q3 In 1660, the Royal Society was founded in London. What was its famous motto encouraging experiments? / 2m

CORE FACT:

EXPLANATION:

Q4 In 1665, what world-first scientific journal was launched by the Royal Society to share discoveries? / 2m

CORE FACT:

EXPLANATION:

Q5 Why was 17th-century English doctor Thomas Sydenham known as the "English Hippocrates"? / 2m

CORE FACT:

EXPLANATION:

Q6 What new idea did Thomas Sydenham introduce about identifying and classifying diseases? / 2m

CORE FACT:

EXPLANATION:

Q7 Which two common childhood diseases did Sydenham prove were separate illnesses for the first time? / 2m

CORE FACT:

EXPLANATION:

Q8 In 1543, which brilliant anatomist published De Fabrica after dissecting human bodies in Padua? / 2m

CORE FACT:

EXPLANATION:

⚠ **EXAMINER WARNING & FATAL PITFALL**

Fatal Examiner Trap: Overstating the immediate practical impact of Vesalius. Vesalius revolutionized anatomical knowledge by proving Galen made over 300 errors, but his discoveries did not improve surgical treatments or cure disease in the 16th century.

KT 2.2: William Harvey's Circulation & The 1665 Great Plague

Key Enquiry Question: "How did William Harvey's discovery of blood circulation transform physiological understanding?"

CHRONOLOGICAL SPINE

1616 Harvey Lectures on Heart → 1628 De Motu Cordis Published → 1649 Harvey's Theory Provoke... → 1661 Malpighi Observes Capill... → c1700 Harvey's Circulation Taught

Q1 In 1628, what landmark book did William Harvey publish in Frankfurt proving blood circulation? / 2m

CORE FACT:

EXPLANATION:

Q2 What royal role did English doctor William Harvey hold under King James I and King Charles I? / 2m

CORE FACT:

EXPLANATION:

Q3 What mistaken belief about blood production had Galen taught for over 1,400 years? / 2m

CORE FACT:

EXPLANATION:

Q4 What mathematical calculation did Harvey make to prove the liver could not be making blood? / 2m

CORE FACT:

EXPLANATION:

Q5 What experiment on a human arm did Harvey carry out using a tight band to prove one-way blood flow? / 2m

CORE FACT:

EXPLANATION:

Q6 In 1665, roughly how many people in London died during the Great Plague? / 2m

CORE FACT:

EXPLANATION:

Q7 What ancient ideas about disease causes persisted during the Great Plague of 1665? / 2m

CORE FACT:

EXPLANATION:

Q8 Under the 1665 Plague Orders, what was done to homes where a family member caught the plague? / 2m

CORE FACT:

EXPLANATION:

⚠ **EXAMINER WARNING & FATAL PITFALL**

Fatal Examiner Trap: Claiming Harvey discovered blood. Harvey discovered the CIRCULATION of blood pumped around the body in a closed loop, disproving Galen's theory that blood was constantly produced by the liver and consumed by the body.

KT 3.1: Germ Theory & Microbe Hunters (c1861–c1883)

Key Enquiry Question: "How did Jenner's development of the smallpox vaccination initiate preventative medicine in the 18th century?"

CHRONOLOGICAL SPINE | 1741 Coram's Foundling Hospital → 1750s Specialist Voluntary Hosp... → 1796 Jenner Inoculates James ... → 1798 Smallpox Vaccine Inquiry → 1852 Compulsory Vaccination ...

Q1 What mistaken theory claimed that rotting matter created living maggots and bacteria out of nowhere? / 2m

CORE FACT:

EXPLANATION:

Q2 In 1861, Louis Pasteur published his Germ Theory. What did he prove was the real cause of decay? / 2m

CORE FACT:

EXPLANATION:

Q3 What famous experiment with swan-neck flasks did Louis Pasteur carry out to prove germs were in the air? / 2m

CORE FACT:

EXPLANATION:

Q4 Which German scientist developed methods to stain, photograph, and grow specific bacteria on agar jelly? / 2m

CORE FACT:

EXPLANATION:

Q5 Which two deadly human disease bacteria were discovered and isolated by Robert Koch in 1882 and 1883? / 2m

CORE FACT:

EXPLANATION:

Q6 What four-step method did Koch introduce to prove a specific microbe caused a specific disease? / 2m

CORE FACT:

EXPLANATION:

Q7 What weakened vaccines did Louis Pasteur create after discovering how germs work? / 2m

CORE FACT:

EXPLANATION:

Q8 How did war between France and Germany (1870–71) speed up discoveries by Pasteur and Koch? / 2m

CORE FACT:

EXPLANATION:

⚠ **EXAMINER WARNING & FATAL PITFALL**

Fatal Examiner Trap: Confusing Pasteur and Koch's contributions. Pasteur discovered Germ Theory (microbes cause decay and disease) and developed rabies vaccines; Koch identified SPECIFIC bacteria (anthrax, TB, cholera) using chemical dyes and agar.

KT 3.2: Prevention & Hospital Care: Smallpox Vaccine & Nightingale

Key Enquiry Question: "How did the breakthroughs of Pasteur, Koch, and Ehrlich establish the Germ Theory of disease?"

CHRONOLOGICAL SPINE

1861 Pasteur's Germ Theory → 1876 Koch Identifies Anthrax → 1882 Koch Stains Tuberculosis ... → 1883 Koch Identifies Cholera → 1909 Ehrlich's Salvarsan 606

Q1 Before vaccination, what dangerous method involved scratching live smallpox scab pus into the skin? / 2m

CORE FACT:

EXPLANATION:

Q2 In 1721, who brought the Turkish practice of smallpox inoculation to Britain? / 2m

CORE FACT:

EXPLANATION:

Q3 In the 1790s, what country observation did Edward Jenner investigate regarding dairy milkmaids? / 2m

CORE FACT:

EXPLANATION:

Q4 In May 1796, what famous experiment did Jenner perform on eight-year-old James Phipps? / 2m

CORE FACT:

EXPLANATION:

Q5 Why did professional inoculators and private doctors oppose Jenner's free smallpox vaccine? / 2m

CORE FACT:

EXPLANATION:

Q6 In 1854, what war prompted Florence Nightingale to take 38 nurses to Scutari hospital? / 2m

CORE FACT:

EXPLANATION:

Q7 When Nightingale arrived at Scutari in 1854, what was the shocking death rate among wounded soldiers? / 2m

CORE FACT:

EXPLANATION:

Q8 What sanitary changes did Nightingale make that cut the death rate at Scutari to under 2%? / 2m

CORE FACT:

EXPLANATION:

⚠ **EXAMINER WARNING & FATAL PITFALL**

Fatal Examiner Trap: Stating Jenner knew how vaccines worked. Jenner proved empirically that cowpox protected against smallpox, but he had no idea why because Germ Theory was not discovered until 65 years later in 1861.

KT 3.3: The Surgical Revolution & Public Health (1847–1875)

Key Enquiry Question: "How did Simpson's chloroform and Lister's carbolic acid overcome surgical pain and infection?"

CHRONOLOGICAL SPINE

1846 Liston Amputates with Et... → 1847 Simpson Discovers Chlor... → 1853 Queen Victoria Inhales C... → 1867 Lister's Carbolic Spray → 1890s Aseptic Theatre Revolution

Q1 What three deadly problems limited early 19th-century surgery to quick amputations? / 2m

CORE FACT:

EXPLANATION:

Q2 In 1847, which Scottish doctor discovered the powerful anaesthetic effects of chloroform? / 2m

CORE FACT:

EXPLANATION:

Q3 In 1853, what royal event stopped religious objections to using chloroform in childbirth? / 2m

CORE FACT:

EXPLANATION:

Q4 What was the "Black Period of Surgery" that happened after anaesthetics were first used c1846–1870? / 2m

CORE FACT:

EXPLANATION:

Q5 In 1865, which surgeon used carbolic acid on compound fractures after reading Pasteur's Germ Theory? / 2m

CORE FACT:

EXPLANATION:

Q6 In 1831, what water-borne disease first hit Britain, killing over 30,000 people? / 2m

CORE FACT:

EXPLANATION:

Q7 What government policy of "leave alone" meant councils did not want to spend taxes on sewers? / 2m

CORE FACT:

EXPLANATION:

Q8 In 1842, who published a landmark report proving dirty living conditions caused poverty and disease? / 2m

CORE FACT:

EXPLANATION:

⚠ **EXAMINER WARNING & FATAL PITFALL**

Fatal Examiner Trap: Believing anaesthetics immediately made surgery safer. Chloroform created a "Black Period" of surgery (1846–1870) where surgeons attempted longer, deeper operations, increasing deaths from infection until Lister introduced antiseptics.

KT 4.1: Modern Medicine: Magic Bullets, Penicillin & The NHS

Key Enquiry Question: "How did Edwin Chadwick, John Snow, and the 1875 Public Health Act conquer cholera and unsanitary towns?"

CHRONOLOGICAL SPINE

1842 Chadwick's Sanitary Report → 1848 First Public Health Act → 1854 Snow Removes Broad St ... → 1858 Great Stink Spurs Bazalg... → 1875 Compulsory Public Healt...

Q1 What was Paul Ehrlich's idea of a "Magic Bullet" in modern medicine? / 2m

CORE FACT:

EXPLANATION:

Q2 In 1909, what first magic bullet was discovered by Paul Ehrlich and Sahachiro Hata to cure syphilis? / 2m

CORE FACT:

EXPLANATION:

Q3 In 1932, what second magic bullet dye was discovered by Gerhard Domagk to cure blood poisoning? / 2m

CORE FACT:

EXPLANATION:

Q4 Who was the Minister of Health in Clement Attlee's Labour government who created the NHS in 1948? / 2m

CORE FACT:

EXPLANATION:

Q5 How did poor working-class families pay for healthcare before the NHS was created in 1948? / 2m

CORE FACT:

EXPLANATION:

Q6 In September 1928, how did Alexander Fleming accidentally discover penicillin at St Mary's Hospital? / 2m

CORE FACT:

EXPLANATION:

Q7 In 1938, which two Oxford scientists began work to purify and extract usable penicillin powder? / 2m

CORE FACT:

EXPLANATION:

Q8 In May 1940, what experiment on mice proved penicillin was a miracle cure for bacterial infection? / 2m

CORE FACT:

EXPLANATION:

⚠ **EXAMINER WARNING & FATAL PITFALL**

Fatal Examiner Trap: Crediting Fleming alone for penicillin. Fleming accidentally discovered the mould in 1928 but could not purify it; Florey and Chain turned it into a usable drug and secured US industrial mass production during WWII.

SCORE: ____ / 16

KT 5.1: The Western Front: Trench Geography & Environmental Illness

Key Enquiry Question: "How did Alexander Fleming, Florey, and Chain discover and mass-produce the wonder drug penicillin?"

CHRONOLOGICAL SPINE

1928 Fleming Observes Penicillin → 1938 Florey & Chain Oxford Re... → 1941 Albert Alexander Clinical ... → 1941 Florey Solicits US Industri... → 1944 D-Day Mass Supply Guar...

Q1 What is the military definition of a "salient", such as the British position at Ypres? / 2m

CORE FACT:

EXPLANATION:

Q2 Why was holding the town of Ypres vital for the British Army throughout the war? / 2m

CORE FACT:

EXPLANATION:

Q3 On 1 July 1916, the British launched the Battle of the Somme. How many casualties did they suffer on Day 1? / 2m

CORE FACT:

EXPLANATION:

Q4 In April 1917, what underground medical facility did British miners build beneath Arras? / 2m

CORE FACT:

EXPLANATION:

Q5 At the Battle of Cambrai in November 1917, what modern weapon was mass deployed for the first time? / 2m

CORE FACT:

EXPLANATION:

Q6 What environmental conditions caused Trench Foot among soldiers on the Western Front? / 2m

CORE FACT:

EXPLANATION:

Q7 State two daily rules ordered by British officers to prevent Trench Foot. / 2m

CORE FACT:

EXPLANATION:

Q8 What surgical operation was carried out for severe, gangrenous Trench Foot? / 2m

CORE FACT:

EXPLANATION:

⚠ **EXAMINER WARNING & FATAL PITFALL**

Fatal Examiner Trap: Describing trenches as simple straight ditches. Trenches were constructed in a zig-zag traverse pattern so an exploding shell or enemy machine gunner could not shoot down the entire length of the trench.

KT 5.2: Battlefield Trauma & Weapons: Shrapnel, Gas & Gangrene

Key Enquiry Question: "How did high-tech medical breakthroughs, DNA discoveries, and the NHS transform modern healthcare in the 20th century?"

CHRONOLOGICAL SPINE | 1948 Bevan Launches Free NHS → 1953 Crick & Watson Model DNA → 1973 Hounsfield Invents CT Sc... → 1978 First IVF Baby Louise Brown → 2003 Human Genome Mapped

Q1 Which weapon caused roughly 58% of all wounds and injuries on the Western Front? / 2m

CORE FACT:

EXPLANATION:

Q2 Why was the farming soil of Flanders so dangerous when soldiers were wounded? / 2m

CORE FACT:

EXPLANATION:

Q3 What deadly wound infection produced gas bubbles and a foul smell in dying muscle? / 2m

CORE FACT:

EXPLANATION:

Q4 From late 1914, what routine injection was given to every wounded British soldier? / 2m

CORE FACT:

EXPLANATION:

Q5 In autumn 1915, what steel helmet was issued to British troops, cutting fatal head wounds by 80%? / 2m

CORE FACT:

EXPLANATION:

Q6 In April 1915, what poisonous gas was first used by the German Army at Second Ypres? / 2m

CORE FACT:

EXPLANATION:

Q7 In late 1915, which colourless gas was introduced that was six times deadlier than chlorine? / 2m

CORE FACT:

EXPLANATION:

Q8 In 1917, what horrific blistering gas was first used by Germany, burning skin through clothes? / 2m

CORE FACT:

EXPLANATION:

⚠ **EXAMINER WARNING & FATAL PITFALL**

Fatal Examiner Trap: Believing poison gas caused the majority of deaths on the Western Front. Artillery shells and shrapnel caused 58% of all battlefield casualties; gas caused less than 3% of deaths, though it caused immense psychological terror.

SCORE: ____ / 16

KT 5.3: The Chain of Evacuation: RAP to Base Hospitals

Key Enquiry Question: "How did the terrain and trench system of the British Sector of the Western Front affect soldiers' health and wounds?"

CHRONOLOGICAL SPINE

Oct 1914 First Ypres Creates Salient → Apr 1915 Second Ypres: Chlorine G... → Jul 1916 Somme: 57,000 Day 1 Los... → Apr 1917 Arras Underground Quarr... → Jul 1917 Passchendaele Quagmire

Q1 What was the main purpose of the "Chain of Evacuation" on the Western Front? / 2m

CORE FACT:

EXPLANATION:

Q2 Where was the Regimental Aid Post (RAP) located on the battlefield? / 2m

CORE FACT:

EXPLANATION:

Q3 What medical care could a doctor provide at a Regimental Aid Post (RAP)? / 2m

CORE FACT:

EXPLANATION:

Q4 Which medical station was located roughly one mile behind the frontline trenches? / 2m

CORE FACT:

EXPLANATION:

Q5 Where were Casualty Clearing Stations (CCS) set up on the Western Front? / 2m

CORE FACT:

EXPLANATION:

Q6 What vital medical sorting system was carried out at the Casualty Clearing Station (CCS)? / 2m

CORE FACT:

EXPLANATION:

Q7 Where were large British Base Hospitals located on the Western Front? / 2m

CORE FACT:

EXPLANATION:

Q8 What dangerous frontline transport job was carried out by female FANY volunteers? / 2m

CORE FACT:

EXPLANATION:

⚠ **EXAMINER WARNING & FATAL PITFALL**

Fatal Examiner Trap: Confusing the stages of the Chain of Evacuation. The precise sequence: Stretcher Bearers → RAP (first aid within 200m) → Dressing Station (triage at 1 mile) → CCS (vital life-saving surgery at 10 miles) → Base Hospital on French coast.

KT 5.4: Surgical Breakthroughs: Thomas Splint, X-Rays & Blood Storage

Key Enquiry Question: "How did the Chain of Evacuation, mobile X-rays, blood transfusions, and surgery transform Western Front medicine (1914–1918)?"

CHRONOLOGICAL SPINE

1914 RAMC Evacuation Chain F... → 1915 Curie Deploys Mobile X-R... → 1915 Sodium Citrate Blood Stor... → 1916 Thomas Splint Slashes M... → Nov 1917 Cambrai: First Blood Depot

Q1 What surgical technique called "debridement" was used to stop wound infections? / 2m

CORE FACT:

EXPLANATION:

Q2 In 1915, what chemical wound-washing method was developed by Alexis Carrel and Henry Dakin? / 2m

CORE FACT:

EXPLANATION:

Q3 What was a major practical problem with the Carrel-Dakin solution on the front line? / 2m

CORE FACT:

EXPLANATION:

Q4 In 1914–15, what was the survival rate for soldiers suffering a fractured femur (thigh bone)? / 2m

CORE FACT:

EXPLANATION:

Q5 In December 1915, who brought the Thomas Splint to the Western Front, training military doctors? / 2m

CORE FACT:

EXPLANATION:

Q6 In 1901, what major discovery about human blood was made by Austrian doctor Karl Landsteiner? / 2m

CORE FACT:

EXPLANATION:

Q7 Which blood group was found in 1907 to be the "universal donor" safe for emergency transfusions? / 2m

CORE FACT:

EXPLANATION:

Q8 In 1915, what chemical did Richard Lewisohn add to blood to prevent it clotting? / 2m

CORE FACT:

EXPLANATION:

⚠ **EXAMINER WARNING & FATAL PITFALL**

Fatal Examiner Trap: Believing blood could be stored at the start of the war. In 1914, blood clotted immediately, requiring direct donor-to-patient transfusion. Only the addition of sodium citrate (1915) and glucose (Rous & Turner, 1916) allowed Oswald Robertson to store blood at Cambrai in 1917.

ENQUIRIES 1–4: ERAS 1 & 2 MEDIEVAL & RENAISSANCE MEDICINE (C1250–C1700) • QS 1–32 | **OFFICIAL 2-MARK STANDARD: 1M FACT + 1M EXPLANATION**

Q1 God was punishing people for personal sins or testing their Christian faith

Illness was seen as punishment from God, meaning healing required prayer, confession, and repentance.

Q2 Monks hand-copied books in monasteries, and universities taught only Church-approved texts

Because the Church ran universities and book copying, questioning Galen was treated as a religious crime (heresy).

Q3 A conjunction of Saturn, Jupiter, and Mars

Believed to create poisonous air that descended to earth as deadly, foul-smelling miasma.

Q4 Bad air corrupted by rotting waste, swamps, corpses, or filth that poisoned the body

Led wealthy people to carry sweet-smelling flower posies and light scented fires to clear the air.

Q5 Publicly whipping themselves with leather scourges tipped with iron spikes

Hoped to stop God's anger and end the plague through voluntary bodily suffering.

Q6 Blood, Phlegm, Yellow Bile (Choler), and Black Bile (Melancholy)

Good health was believed to exist when all four fluids were in natural balance inside the body.

Q7 Hot and Wet, linked with the season of Spring and the element of Air

Too much blood was believed to cause fevers, red skin, and an overly optimistic (sanguine) mood.

Q8 The Theory of Opposites

Treated an excess of one quality with its direct opposite (e.g. cold phlegm treated with hot pepper).

Q9 Bloodletting (phlebotomy)

Aimed to remove excess hot, wet blood to cool down fevers and restore internal humoural balance.

Q10 Opening a vein (with a fleam), Cupping (heated glass cups), and Leeches

Chosen depending on the patient's age, illness, and how physically weak they were.

Q11 Laxatives (to empty bowels) and Emetics (to make patients vomit)

Believed to physically flush poisonous humours and corrupt bile out of the digestive tract.

Q12 The Regimen Sanitatis (Rule of Health)

Originated in Salerno and urged people to balance air, exercise, sleep, diet, excretion, and emotion.

Q13 "Care Not Cure"

Hospitals aimed to provide prayer, warmth, rest, and clean food rather than active medical treatments.

Q14 Monks and nuns, funded through Church charity and wills left by wealthy citizens

Run as religious houses where daily prayer, Christian charity, and quiet rest took first place.

Q15 The University-Trained Physician

Studied for up to 10 years in Latin reading Galen; affordable only for the king and wealthy nobles.

Q16 Between one-third and one-half of the population (approx. 1.5 to 2 million people)

Spread inland rapidly along trade roads, reaching London by autumn 1348 and killing over a third of the country.

Q17 Renaissance Humanism

Encouraged people to test ideas through scientific experiments rather than blindly accepting old books.

Q18 The movable-type printing press (Johannes Gutenberg)

Stopped monk copying mistakes and allowed anatomical books with drawings to be mass-produced across Europe.

Q19 "Nullius in Verba" ("Take nobody's word for it")

Stressed that scientific theories had to be proven by repeatable experiments, not ancient tradition.

Q20 Philosophical Transactions

Allowed scientists and doctors across Europe to read, test, and share new medical experiments.

Q21 He revived careful bedside observation of patients rather than relying on old textbooks

Stressed recording patient symptoms closely at the bedside instead of debating abstract medical theories.

Q22 Argued diseases were distinct species that could be grouped into families like plants

Shifted medicine away from unique humoural imbalances toward identifying specific, recurring illnesses.

Q23 Measles and Scarlet Fever

Showed that illnesses previously lumped together as general fevers had distinct symptoms and rashes.

Q24 Andreas Vesalius

Professor of Surgery at Padua; transformed anatomical science through direct human dissection.

Q25 De Motu Cordis (On the Motion of the Heart and Blood)

Proved mathematically and through experiments that blood circulates continuously in one closed loop.

Q26 Royal Physician to the King

His high royal status gave him money and protection to carry out pioneering experiments on the heart.

Q27 That the liver constantly made new blood from food, which was burned up by the body like fuel

Galen taught that veins carried blood from the liver, while arteries carried separate vital spirit from the heart.

Q28 Calculated the heart pumped 540 pounds of blood an hour—far more than the body's weight

Proved the liver could not possibly make that much blood; the same blood had to be recycled constantly.

Q29 Pushed blood along a vein with a finger, proving one-way valves stop blood flowing backward

Showed that blood in veins flows only toward the heart, blocked by tiny pocket valves from reversing.

Q30 Approximately 100,000 people (around 20% of London's population)

The worst epidemic in England since 1348, causing mass panic across the crowded capital city.

Q31 Punishment from God, bad planet alignments, and poisonous bad air (miasma)

Shows deep continuity; Renaissance doctors still had not discovered bacteria or that fleas spread plague.

Q32 Locked inside for 28 days, watched by guards, with a red cross painted on the door

Doors were painted with a red cross and "Lord Have Mercy on Us", with burials done only after dark.

Q33 The Theory of Spontaneous Generation
Believed bacteria were the result of rotting matter rather than the cause of decay and disease.

Q34 Microscopic living germs (microbes) in the air, disproving Spontaneous Generation
Showed that airborne microbes caused liquids to spoil, laying the foundation for modern medicine.

Q35 Boiled broth in S-shaped flasks; the broth stayed sterile until the glass neck was broken
Showed germs fell into open liquids from dust in the air, rather than appearing out of nothing.

Q36 Robert Koch
Used industrial chemical dyes to stain clear bacteria and grew pure colonies on solid agar jelly.

Q37 Tuberculosis (1882) and Cholera (1883)
Proved definitively that specific microbes caused specific human illnesses, finally disproving miasma.

Q38 Koch's Postulates
A strict scientific standard: isolate the microbe, grow it pure, test in an animal, and retrieve it again.

Q39 Vaccines for Chicken Cholera, Anthrax (1881), and Rabies (1885)
Discovered that weakened or aged pathogens stimulated protective antibodies without killing the host.

Q40 National rivalry led both governments to pour huge funding into rival medical research teams
Each country competed to prove scientific superiority and discover cures to save soldiers' lives.

Q41 Inoculation (variolation)
Patients caught a mild dose of real smallpox, but risked dying or triggering a full epidemic.

Q42 Lady Mary Wortley Montagu
Had her children inoculated after seeing it in Turkey, popularizing it among the British aristocracy.

Q43 Milkmaids who caught mild cowpox never caught the deadly smallpox virus
Jenner realized that catching mild cowpox gave people lifelong biological protection against smallpox.

Q44 Infected him with cowpox pus from a milkmaid, then exposed him to smallpox; he did not get sick
James developed mild cowpox but completely resisted smallpox infection, proving vaccination worked.

Q45 They stood to lose large profits made by charging wealthy families high fees for inoculation
Money and self-interest led many private doctors to publish pamphlets attacking vaccination.

Q46 The Crimean War (1853–1856)
Newspaper reports of shocking hospital neglect and dying British soldiers caused national outrage.

Q47 Approximately 42% (over 4 in 10 patients died)
Soldiers were dying from typhus, cholera, and dysentery caused by blocked sewers beneath hospital wards.

Q48 Cleaned sewers, washed bedding, scrubbed wards, boiled drinking water, and opened windows
Dramatic proof that cleanliness, fresh air, and proper sanitation saved lives in hospitals.

Q49 Pain (shock), Infection (gangrene), and Bleeding (blood loss)
Surgeons had to amputate limbs in under a minute while conscious patients were held down by assistants.

Q50 James Young Simpson
Tested chloroform fumes with his assistants in Edinburgh, waking up safely under the dinner table.

Q51 Queen Victoria used chloroform during the birth of Prince Leopold
Silenced Church critics who claimed the Bible demanded women feel pain during childbirth.

Q52 Death rates rose because surgeons operated deeper and slower in dirty, unsterilized conditions
Surgeons wore bloodstained coats and didn't wash hands; longer operations carried more deadly bacteria deep inside.

Q53 Joseph Lister
Understood that wound rotting was caused by living airborne germs, not chemical reactions.

Q54 Asiatic Cholera (striking the port of Sunderland)
Caused violent diarrhoea, vomiting, dehydration, and blue skin, often killing people within hours.

Q55 "Laissez-faire" (leave alone)
Local taxpayers and councils refused to pay higher taxes to clean streets or lay water pipes.

Q56 Edwin Chadwick
Used statistics to prove overcrowding, uncollected sewage, and bad water caused widespread illness.

Q57 A chemical drug that seeks out and destroys bacteria without harming healthy human cells
Replaced harsh chemicals like carbolic acid with targeted drugs that killed only specific germs.

Q58 Salvarsan 606 (an arsenic compound)
Tested over 600 chemical compounds before finding that number 606 killed the syphilis bacteria.

Q59 Prontosil (the first sulpha drug)
A bright red dye that killed streptococcus bacteria, saving Domagk's daughter from an arm amputation.

Q60 Aneurin (Nye) Bevan
Overcame fierce opposition from private doctors by allowing hospital consultants to keep private beds.

Q61 Had to pay out of their own pocket, use charity hospitals, or go without treatment
The 1911 National Insurance Act covered only working men, leaving wives and children unprotected.

Q62 Returned from holiday to find penicillium mould had dissolved staphylococcus bacteria on an agar dish
Realized the green mould produced a natural chemical juice that killed dangerous bacteria.

Q63 Howard Florey and Ernst Chain
Revived Fleming's forgotten paper and extracted stable penicillin powder in their Oxford lab.

Q64 Injected eight mice with lethal bacteria; the four given penicillin survived while the other four died
Proved definitively that penicillin worked inside a living body as a powerful antibiotic.

Q65 A battlefield bulge surrounded on three sides by the enemy

Left British troops exposed to German artillery fire from higher ground on surrounding ridges.

Q66 It protected the English Channel ports used for supplies and reinforcements

If Ypres fell, the German army could cut off supply links between Britain and the front line.

Q67 57,000 casualties (over 19,000 killed)

The worst single day in British military history, completely overwhelming frontline medical stations.

Q68 An underground hospital in chalk tunnels (Thompson's Cave)

Contained 700 beds, operating rooms, water, and power, completely safe from shellfire.

Q69 Massed deployment of over 450 British tanks

Broke German lines and created the conditions for the first operational blood bank.

Q70 Standing for days in cold, waterlogged mud and tight wet boots

Slowed blood flow, causing feet to go numb, blister, swell, and turn gangrenous.

Q71 Rubbing feet with whale oil and changing into dry socks twice daily

Officers held daily foot inspections; soldiers worked in pairs to dry each other's feet.

Q72 Amputation of toes or the whole foot

Necessary to stop spreading gangrene and fatal blood poisoning once tissue died.

Q73 High-explosive artillery shells and shrapnel

Exploding steel shell casings threw jagged fragments that shattered bones and ripped muscle.

Q74 The soil had been heavily manured for centuries, filling it with deadly bacteria

Manured soil contained tetanus and gas gangrene bacteria that multiplied in deep wounds without oxygen.

Q75 Gas Gangrene

Soil bacteria produced poisons that turned muscle black, killing tissue within hours and causing death in a day.

Q76 Anti-tetanus serum (ATS)

Massively reduced deaths from tetanus (lockjaw) caused by soil bacteria entering open wounds.

Q77 The Brodie Helmet

Made of pressed steel with a wide rim to deflect falling shrapnel and dirt.

Q78 Chlorine Gas

A greenish-yellow choking gas that damaged the lungs, causing soldiers to drown in bodily fluids.

Q79 Phosgene Gas

Smelled like mouldy hay and acted slowly, often killing an exposed soldier within 48 hours.

Q80 Mustard Gas

An oily liquid that stayed in the soil for weeks, causing severe skin blisters, blindness, and lung burns.

Q81 To move, sort, and treat wounded soldiers in stages from the battlefield to safety

Ensured casualties were treated quickly without clogging up frontline medical stations.

Q82 Within 200 metres of the frontline trench, in a dugout or ruined cellar

Positioned close to the fighting so wounded men could receive basic first aid within minutes.

Q83 Basic first aid (bandages, pain relief, splints); no surgery could be done

Patched up walking wounded to fight, or sent stretcher cases back to dressing stations.

Q84 The Dressing Station (Advanced or Main Dressing Station)

Run by the Field Ambulance; gave fresh dressings, tetanus jabs, and hot drinks.

Q85 7 to 12 miles behind the front line, near railways and outside shell range

Positioned near railheads so stabilized wounded could be sent by train to coastal hospitals.

Q86 Triage: sorting wounded into walking cases, urgent surgery, and dying

Focused limited surgeons and operating tables on men with the best chance of survival.

Q87 Near French and Belgian ports (Calais, Boulogne, Le Havre, Rouen)

Close to the sea so long-term casualties could be loaded directly onto hospital ships for Britain.

Q88 Driving motor ambulance convoys under shellfire between medical stations

FANY drivers were the first women to drive British military vehicles on the Western Front.

Q89 Cutting away all dead, damaged, and infected tissue around the wound

Removed the dead flesh that dangerous gas gangrene bacteria needed to grow.

Q90 The Carrel-Dakin Method

Flushed a mild antiseptic bleach solution through rubber tubes directly into deep wounds.

Q91 The chemical solution spoiled quickly and had to be freshly mixed every 6 hours

Hard to prepare during large battles when thousands of casualties arrived at once.

Q92 Only 20% survived (an 80% death rate)

Bone ends ground together in bumpy wagons, tearing large blood vessels and causing fatal shock.

Q93 Robert Jones (nephew of Hugh Owen Thomas)

Set up training centres for medics; survival rates for femur fractures jumped from 20% to 82%.

Q94 Discovered different human blood groups (A, B, and O)

Showed why earlier transfusions often caused blood to clump together and kill the patient.

Q95 Blood Group O

Essential for battle casualties because Group O blood could be given safely without testing.

Q96 Sodium Citrate

Stopped blood clotting without poisoning the wounded patient receiving the transfusion.